



Fabian Karl

PhD-Candidate

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EMPLOYMENT HISTORY

10/25 – Present Munich, Germany	PhD-Candidate Technical University of Munich
09/24 – 09/25 Ulm, Germany	PhD-Candidate Ulm University

EDUCATION

07/22 – 09/24 Ulm, Germany	M.Sc. Computer Science Ulm University Grade: 1.2 Thesis title: Retrieval Augmented Information Extraction: Enhancing Language Models for Extracting Bibliographic Metadata from Heterogeneous Web Sources with CRAWLDoc.
10/19 – 07/22 Ulm, Germany	B.Sc. Computer Science Ulm University Grade: 1.3 Thesis title: Transformers are Short Text Classifiers: A Study of Inductive Short Text Classifiers on Benchmarks and Real-world Datasets.
09/16 – 09/19 Ehingen, Germany	German Abitur Technisches-Gymnasium Ehingen Grade: 1.2

LANGUAGES

German	● ● ● ● ●	English	● ● ● ● ●
French	● ● ● ● ●		

RESEARCH PUBLICATIONS

2025	HYDRA: A Multi-Head Encoder-only Architecture for Hierarchical Text Classification Association for Computational Linguistics Karl, F., & Scherp, A. (2025, November). HYDRA: A Multi-Head Encoder-only Architecture for Hierarchical Text Classification. In <i>Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing</i> (pp. 9303-9314).
2025	Crawldoc: A dataset for robust ranking of bibliographic documents SCOLIA F. Karl and A. Scherp, "Crawldoc: A dataset for robust ranking of bibliographic documents", SCOLIA@ECIR 2025.

2024	Continual Learning for Encoder-only Language Models via a Discrete Key-Value Bottleneck  arXiv e-prints A. Diera, L. Galke, F. Karl, and A. Scherp, "Continual learning for encoder-only language models via a discrete key-value bottleneck" DOI: 10.48550/ARXIV.2412.08528. arXiv: 2412.08528.
2023	GenCodeSearchNet: A benchmark test suite for evaluating generalization in programming language understanding  Association for Computational Linguistics A. Diera, A. Dahou, L. Galke, F. Karl, F. Sihler, and A. Scherp, "GenCodeSearchNet: A benchmark test suite for evaluating generalization in programming language understanding" GenBench Workshop@EMNLP 2023, DOI: 10.18653/v1/2023.genbench-1.2.
2022	Are We Really Making Much Progress in Text Classification? A Comparative Review  arXiv e-prints L. Galke, A. Diera, B. X. Lin, et al., "Are We Really Making Much Progress in Text Classification? A Comparative Review" DOI: 10.8550/arXiv.220.0395. arXiv: 220.0395.
2022	Transformers are Short-Text Classifiers  Springer F. Karl and A. Scherp, "Transformers are Short-Text Classifiers" CD-MAKE 2023, Springer, ISBN: 978-3-031-0836-6. DOI: 10.1007/978-3-031-0837-3_7.

★ INVITED TALKS

2023	Transformers are short-text classifiers Presented for Ernst & Young (EY) Germany R&D.
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CERTIFICATES

2025	AI Agents Fundamentals  Awarded by Hugging Face
2023	Effective MLOps - Model Development  Awarded by Weights & Biases.

CODING

• Python	• PyTorch	• Transformers
• Java	• Git	• LaTeX